Bahati Patrick

Data Scientist | AI & Automation | SAP ABAP

bahatipatrick87@gmail.com · linkedin.com/in/bahatipatrick87 · github.com/bahatipatrick87 · personal-website-hnyb.vercel.app · Messina, Italy

PROFILE SUMMARY

Data Scientist and AI Automation Developer with hands-on experience in machine learning, deep learning, big data engineering, AI-powered business automation, and modern SAP development. Currently completing a Master's Degree in Data Science at the Università degli Studi di Messina, with research experience in spiking neural networks and computational neuroscience. Practical training in AI & Business Automation and modern SAP ABAP Cloud development, including SAP BTP, CDS, RAP, OData V4, and Clean Core principles. Comfortable working across technical and business environments and communicating with both technical and non-technical stakeholders.

EDUCATION

Master's Degree in Data Science — **2024 – 2026**
Università degli Studi di Messina — **Messina, Italy**

Bachelor's Degree in Business Information Technology — **2020 – 2023**
Nkumba University — CGPA: 4.45 / 5.00 — **Entebbe, Uganda**

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — **Oct 2026**
DeepLearning.AI & Stanford Online (Andrew Ng) — Coursera

- Completed the first course of the Machine Learning Specialization: linear and logistic regression, gradient descent, regularization, and classification.
- Verified at coursera.org/verify/Q2VYSO6M7Q6N.

AI & Business Automation — **Sept 2026**
Academy Rapido — Ready for IT programme by Fondazione Italiana Accenture & The Human Safety Net

- Built AI-powered business automation workflows using n8n, OpenAI, Google Gemini, REST APIs, and Google Workspace.
- Applied prompt engineering, RAG pipelines, AI agents, and multi-agent workflow design to business automation projects.

SAP ABAP Cloud Development — **2026**
Training & Practical Projects

- Completed practical training in modern SAP development using ABAP Cloud, SAP BTP, CDS Views, RAP, OData V4, and Clean Core.
- Developed sample ABAP applications using modern ABAP syntax and object-oriented programming concepts.
- Created semantic CDS Views for business data modeling and built a RAP Business Object with data modeling, behavior definition, and service exposure.
- Configured service bindings to expose RAP applications through OData V4 and applied Clean Core principles for cloud-ready SAP extensions.
- Gained practical knowledge of the SAP BTP ABAP Environment (Steampunk) and the backend concepts supporting SAP Fiori applications.

Next.js App Router Fundamentals — **Aug 2025**
Vercel — Certificate of Completion

- Hands-on training in Next.js App Router: server components, routing, data fetching, and full-stack deployment.

WORK EXPERIENCE

Computational Neuroscience Research Intern — **March 2026 – June 2026**
Università degli Studi di Messina — Brian Lab — **Messina, Italy**

- Implemented biologically inspired Spiking Neural Networks (SNNs) using the Brian2 simulator.
- Developed and simulated Leaky Integrate-and-Fire (LIF) and conductance-based neuron models.
- Designed Spike-Timing-Dependent Plasticity (STDP) and Reward-Modulated STDP (R-STDP) learning systems.
- Investigated excitatory/inhibitory synaptic dynamics and eligibility trace mechanisms; visualized membrane voltage, synaptic conductance, and adaptive weight changes (LTP/LTD).

Business Coach — **October 2023 – October 2024**
AVSI Foundation — **Isingiro District, Nakivale, Uganda**

- Mentored micro and large enterprises using data-informed business growth strategies and market analysis.
- Collaborated with private-sector employers on job placement, reporting labor market data for workforce development.

Data Analyst Intern — **June 2022 – September 2022**
Stratcom Communications and Computer Solutions — **Uganda**

- Cleaned, processed, and analyzed datasets to support data-driven business decisions.
- Developed dashboards and automated reporting solutions using Python and BI tools.
- Supported system performance monitoring and optimization through real-time data analysis.

Data Clerk & Research Assistant — **2021 – 2022**
Windle International Uganda — **Uganda**

- Managed large-scale field data collection, validation, and database maintenance for education-focused research.
- Performed quantitative and qualitative analysis while ensuring data accuracy and research compliance.

Community Facilitator **2019 – 2020**
DRDIP **Isingiro District, Uganda**

- Facilitated stakeholder–community communication and delivered capacity-building programs with outcome monitoring.

SELECTED PROJECTS

Federated Learning: Comparing FedAvg vs. FedMedian Aggregation

Python, PyTorch, Flower — **GitHub**

- Simulated federated learning with 8 non-IID edge clients on CIFAR-10 using a custom residual CNN, comparing FedAvg and FedMedian aggregation and reaching 82% test accuracy.
- Designed a Byzantine-fault test (one corrupted client): FedMedian retained 64% accuracy vs. FedAvg's collapse to 10%.

Reward-Modulated Spiking Neural Network (R-STDP)

Brian2, Python

- Designed a conductance-based LIF neural network with excitatory and inhibitory synapse populations.
- Implemented STDP and reward-modulated synaptic plasticity with eligibility traces and visualized learning dynamics.

Plant Disease Detection Using CNN

TensorFlow, Keras, Python

- Built a CNN-based model for multi-class plant disease classification on the PlantVillage dataset (70,000+ images, 38 classes).
- Applied data augmentation, dropout, batch normalization, and early stopping, achieving ~91% test accuracy.

Real-Time Credit Card Fraud Detection Pipeline

PySpark, Kafka, Hadoop, Docker, Streamlit

- Architected a Lambda Architecture pipeline processing 280,000+ transactions using Spark Structured Streaming and Kafka (sub-2-second latency) with Hadoop MapReduce batch analytics.
- Trained Random Forest and Logistic Regression models (Spark MLlib) served through a containerized Streamlit dashboard.

Big Data Analytics Final Project — Apache HBase & FDA CAERS Dataset

Hadoop, HBase, Docker, Python — University Project

- Deployed a multi-node Hadoop/HBase cluster (HDFS, YARN, ZooKeeper) with Docker and built a Python ETL pipeline converting 149,945 FDA adverse-event reports (233,821 product records) into HBase via MapReduce ImportTsv.
- Designed a normalized HBase schema (namespaces, column families, versioning, compression) and implemented DDL, CRUD/DML, and cluster administration via the HBase Shell.

AI Customer Support Automation Platform

n8n, OpenAI, Supabase, Airtable, Zendesk

- Designed an AI support workflow with semantic search, ticket classification, Airtable logging, and Zendesk escalation.
- Built a RAG chatbot using a Supabase Vector Store and HTML/CSS analytics dashboards.

Multi-Agent Marketing Campaign Generator

n8n, Google Gemini, Google Docs

- Developed a multi-agent AI system for campaign planning, content generation, and quality review.
- Automated brand strategy and editorial plan generation with consistent tone enforcement through Google Docs integration.

TECHNICAL SKILLS

Programming	Python, SQL, Java, JavaScript, HTML/CSS	**SAP Development**	SAP ABAP Cloud, SAP BTP, CDS Views, RAP, OData V4, Clean Core, Fiori concepts
Machine Learning	Supervised learning (regression & classification), Scikit-learn, TensorFlow, Keras, PyTorch, Spiking Neural Networks (Brian2), STDP, R-STDP	**Federated Learning**	Flower, non-IID data partitioning, FedAvg, FedMedian, Byzantine fault tolerance
Scientific Computing	NumPy, Pandas, SciPy, simulation modeling, feature engineering	**Big Data**	Spark (PySpark, Spark SQL), Kafka, Hadoop (HDFS, MapReduce), HBase, Hive
Databases	MySQL, PostgreSQL, MongoDB, SQL Server, Oracle	**Data Visualization**	Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau, Excel
Computer Vision	OpenCV, CNNs for image classification	**AI & Automation**	n8n, OpenAI, Gemini, RAG, AI agents, multi-agent workflows, prompt engineering, REST APIs
Web & Backend	React, Angular, Node.js, Next.js, Spring Boot, ASP.NET Core, Django	**DevOps & Tools**	Docker, Docker Compose, Kubernetes, Git, GitHub
Research Methods	Data preprocessing, statistical analysis, academic writing, experimental design		

LANGUAGES

English — Full Professional **Kinyarwanda** — Full Professional **Luganda** — Full Professional **Swahili** — Intermediate

Italian — Intermediate

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